Nth-to-Default CDS - Trading, Pricing & Risk

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Abstract

In this article we provide a quantitative overview of of how to price Nth-to-Default Credit Default Swaps within the standard one-factor Gaussian copula framework widely used in credit derivatives markets. The document describes the structure and economic intuition of the instrument, outlines the relationship between basket spreads, single-name credit risk, and default dependence, and presents practical approaches for valuation. In particular, we discuss analytical and Monte Carlo implementations of the one-factor model, methods for combining marginal default probabilities to obtain joint default distributions, and the role of implied correlation calibrated from liquid credit index tranche markets. The note also examines key risk measures—including CS01, correlation delta, and cross-gamma—and discusses practical hedging considerations and market dynamics such as correlation spikes and liquidity stress during credit spread widening.

1 Overview

The Nth-to-Default CDS (NTD) is a credit derivative that references a basket of N bonds, where the protection payment is triggered if n bonds default. The First-to-Default CDS (FTD) is the most liquid of the Nth-to-Default CDS products and the most actively traded. Pricing and risk management requires careful modeling of single-name credit spreads and joint default dependence. In what follows, this article outlines:

- The product and trading rationale,
- Valuation approaches for vanilla, first-to-default, and general Nth-to-Default CDS,
- Correlation modeling and its impact on spreads,
- Risk exposures including correlation delta and cross gamma,
- Hedging strategies and gap risk considerations.

2 The Product

A Nth-to-Default Credit Default Swap is a basket credit derivative referencing a portfolio of N bonds. The instrument transfers the credit risk associated with multiple reference bonds, with the protection payment triggered when the n-th default occurs within the basket during the life of the contract.

Let τ_i denote the default time of bond i, for $i = 1, \ldots, N$. The time of the n-th default in the basket can be expressed using the order statistic

$$\tau_n = \inf\left\{ t \geq 0 : \sum_{i=1}^{N} \mathbf{1}_{\{\tau_i \leq t\}} \geq n \right\}.$$

The contract therefore terminates when τ_n occurs prior to maturity T. At that point, the protection buyer receives a payment corresponding to the loss on the defaulted reference bond, typically equal to

$$(1 - RR) \times \text{Notional},$$

where RR denotes the assumed recovery rate. Premium payments are made periodically by the protection buyer until either the n-th default occurs or the contract reaches maturity.

The most actively traded structure within this class of products is the First-to-Default (FTD) CDS, corresponding to $n = 1$. In this case the protection payment is triggered by the earliest default in the basket

$$\tau_1 = \min(\tau_1, \tau_2, \ldots, \tau_N).$$

More generally, higher order NTD structures allow exposure to the occurrence of multiple defaults within a portfolio. For example, a Second-to-Default contract pays when the second bond in the basket defaults, while a Third-to-Default contract pays upon the third default, and so forth.

NTD CDS instruments are typically used for several purposes in credit markets. They provide a mechanism for yield enhancement, allowing investors to earn higher spreads than those available from single-name CDS in exchange for assuming exposure to correlated defaults within a basket. They are also used as a form of basket CDS protection, allowing investors to hedge credit exposure across a diversified set of bonds. In addition, these products are widely used in correlation trading and dispersion strategies, where market participants express views on the degree of default dependence between credit names.

From a pricing perspective, the value of an NTD CDS is driven primarily by two factors. The first is the level of credit spreads of the underlying bonds, which determines the marginal default probabilities implied by the single-name CDS market. The second is the degree of default dependence between the bonds in the basket, typically represented through an implied correlation parameter linked to a latent macro credit risk factor. The interaction between these two components—individual credit risk and joint default behaviour—forms the central modelling challenge in NTD valuation.

3 Why Trade Nth-to-Default CDS?

Nth-to-Default (NTD) CDS are portfolio credit derivatives that allow investors to express views on **default clustering** and correlation among multiple issuers. By referencing a basket of bonds, these instruments provide exposures that cannot be achieved with single-name CDS alone. Traders and portfolio managers use NTD CDS to manage portfolio risk, speculate on credit correlation, and enhance yield through multi-name strategies. The choice of which n to target — first, second, or higher — drives both the payoff profile and sensitivity to correlation, shaping both trading and hedging strategies.

3.1 First-to-Default

The First-to-Default CDS (FTD) is the most commonly traded NTD instrument due to its liquidity and relative simplicity. It triggers a protection payment upon the first default in the basket, making

it particularly useful for hedging concentrated credit exposure or expressing views on early default risk.

Key features include:

- Generally more liquid than higher-order NTD CDS, facilitating efficient execution,
- Provides a short-correlation exposure, as buyers of FTD protection benefit when defaults are dispersed rather than clustered,
- Allows portfolio managers to hedge concentrated risk in baskets of correlated bonds while maintaining manageable position sizes.

3.2 Nth-to-Default ($n > 1$)

Higher-order NTD CDS enable investors to target scenarios where **multiple defaults occur**. These instruments are inherently more sensitive to correlation and default clustering, making them powerful tools for yield enhancement, structured portfolio strategies and bespoke risk management.

Their applications include:

- Targeted exposure to multi-default scenarios in a basket of bonds, capturing tail risk,
- Long-correlation positions, as higher n increases sensitivity to default clustering,
- Useful in synthetic portfolio structures, bespoke hedging, and correlation trading where a nuanced view of joint defaults is required.

4 Pricing Vanilla CDS

Before tackling multi-name or NTD products, it is essential to understand **single-name CDS pricing**, which forms the foundation for more complex structures. A vanilla CDS consists of a **protection leg**, which compensates for credit events, and a **premium leg** representing periodic payments by the protection buyer. Pricing involves computing the expected present value of these cash flows using hazard rates derived from market-observed single-name CDS spreads.

Mathematically, the present values (PVs) of CDS trade legs are given by:

$$\mathrm{PV}_{\text{protection}} = \sum_{t_i \leq T} \mathbb{E}[(1 - RR)\mathbf{1}_{\tau_i \in [t_{i-1}, t_i]}] \cdot P(0, t_i),$$

$$\mathrm{PV}_{\text{premium}} = \sum_{t_i \leq T} S \cdot \Delta t_i \cdot \mathbb{E}[\mathbf{1}_{\tau_i > t_i}] \cdot P(0, t_i),$$

where S is the par spread, RR the recovery rate, and $P(0, t)$ the discount factor to time 0. Solving for S such that $\mathrm{PV}_{\text{protection}} = \mathrm{PV}_{\text{premium}}$ yields the **par CDS spread**, which is the market-consistent price for entering the protection contract.

5 Pricing First-to-Default CDS

5.1 One-Factor Gaussian Copula - Numerical Integration Approach

The one-factor Gaussian copula (1F GC) framework is widely used in the pricing of Nth-to-Default (NTD) CDS due to its tractability and alignment with market conventions. Its primary advantage

lies in its ability to capture correlated defaults across a basket of bonds using a single latent systemic factor, which simplifies calibration and computation while remaining consistent with single-name CDS spreads. Analytical or semi-analytical solutions, such as numerical integration over the systemic factor, provide fast and stable pricing, making the method suitable for real-time desk applications and sensitivity analysis. Moreover, the model naturally produces an implied correlation parameter that can be calibrated from liquid credit index tranches, facilitating consistency with observable market data.

However, the 1F Gaussian copula also has limitations. Its single-factor structure assumes that all pairwise correlations are identical (equi-correlation), which may misrepresent portfolios with heterogeneous or sector-specific dependencies. The Gaussian assumption implies thin tails, potentially underestimating extreme joint default events and tail risk. Additionally, the model does not explicitly capture time-varying correlations or dynamic contagion effects, and the use of numerical integration or Monte Carlo is still required for higher-order NTD structures, which can become computationally expensive for large baskets. As a result, while the 1F GC is the market standard for many applications, careful risk management and consideration of model limitations are essential when using it for portfolio-level decision-making.

Valuing NTD CDS requires modelling the joint distribution of default times across a basket of bonds. A widely used and market-standard approach is the one-factor Gaussian copula, which captures default dependence via a single latent systemic credit factor while remaining consistent with single-name CDS spreads. This framework separates the influence of macroeconomic or market-wide credit conditions from idiosyncratic bond-specific risk, allowing tractable computation of joint survival probabilities.

Credit Risk Factor: In this model, each bond's default is driven by an underlying latent variable X_i that combines both the systemic credit factor and the bond-specific idiosyncratic component. Conditional on a fixed value of the systemic factor, defaults are independent, which simplifies the calculation of basket survival probabilities. Formally, we define: $X_i = \sqrt{\rho}M + \sqrt{1-\rho}\epsilon_i$, where $M \sim \mathcal{N}(0,1)$ represents a systemic factor and ϵ_i the idiosyncratic risk factor for the individual bond.

Conditional survival: Conditional survival refers to the probability that a bond has not defaulted by time t given a particular realization of the common systemic credit factor in a one-factor Gaussian copula model. Conditioning on M renders the defaults of all bonds independent, allowing tractable computation of joint survival probabilities for the basket under a fixed state of the systemic factor.

Conditional on (M=m), the survival probability of bond (i) is

$$Q_i(t \mid m) = \Phi\Big(\frac{\Phi^{-1}(Q_i(t)) - \sqrt{\rho}m}{\sqrt{1-\rho}}\Big),$$

and the conditional survival probability of the basket for a First-to-Default CDS is

$$\mathbb{P}(\tau_1 > t \mid m) = \prod_{i=1}^{N}(1 - Q_i(t \mid m)).$$

5.1.1 Unconditional survival:

Unconditional survival is the probability that a bond or the basket of bonds survives up to time t without any defaults, integrating over all possible outcomes of the systemic factor. In the one-

factor Gaussian copula framework, the unconditional survival probability of bond i is obtained by integrating the conditional survival over the distribution of M, which is typically standard normal as follows,

$$\mathbb{P}(\tau_1 > t) = \int_{-\infty}^{\infty} \prod_{i=1}^{N} (1 - Q_i(t \mid m))\ \phi(m)\ dm$$

5.2 One-Factor Gaussian Copula - Monte Carlo Approach

The one-factor Gaussian copula Monte Carlo approach provides a flexible numerical method to value Nth-to-Default (NTD) CDS when analytical integration is challenging, particularly for larger baskets or complex payoff structures. In this framework, the default time of each bond is driven by a latent variable $X_i = \sqrt{\rho}, M + \sqrt{1-\rho}, \epsilon_i$, where M represents the common systemic credit factor and ϵ_i are independent idiosyncratic shocks. Correlation among the bonds is introduced entirely through their shared dependence on M. In a Monte Carlo simulation, one repeatedly samples M and the ϵ_i, computes X_i, and maps these latent variables to default times τ_i using the inverse cumulative distribution of the single-name marginal default probabilities.

Conditional on a simulated M, bond defaults are independent, and the simulation allows straightforward counting of the n-th default τ_n in each scenario. Cash flows for the NTD CDS—protection payments upon the n-th default and premium payments up to that time—are discounted to present value for each scenario. The Monte Carlo estimate of the NTD CDS price is obtained by averaging the discounted cash flows across all simulated paths. This method is particularly useful for validating analytical approximations, incorporating complex recovery assumptions, or exploring tail risk scenarios in baskets with heterogeneous credit qualities.

Steps:

1. Sample M and ϵ_i,
2. Construct X_i and map to τ_i via inverse CDF,
3. Determine τ_1, discount cash flows and average in scenarios.

5.3 Standard Monte Carlo with Correlation Matrix

An alternative to the one-factor Gaussian copula is to perform a standard Monte Carlo simulation that directly incorporates the correlation structure of the underlying bonds through a specified correlation matrix. In this approach, the joint distribution of default times is modelled using a multivariate normal vector whose covariance is calibrated to reflect the desired pairwise dependencies between the bonds. This allows greater flexibility than the one-factor model, as it can accommodate arbitrary correlation patterns, including heterogenous or non-equi-correlated exposures across the portfolio.

The simulation proceeds by generating correlated standard normal random variables via Cholesky decomposition of the correlation matrix or, in some cases, using principal component analysis to ensure positive semi-definiteness. These correlated variables are then mapped through the inverse of the marginal default probability distributions to produce simulated default times for each bond. By repeating this procedure over a large number of scenarios, one can estimate both the distribution of the Nth-to-Default time and the corresponding expected cash flows of the CDS. This approach provides a flexible and robust method for pricing NTD and FTD contracts, validating analytical results, and exploring complex basket structures or non-standard correlation assumptions.

Steps:

1. Specify the correlation matrix Σ (equi-correlation often used),
2. Apply Cholesky decomposition $\Sigma = LL^\top$,
3. Sample standard normals Z, then $Y = LZ$,
4. Map to default times using single-name marginals.

6 Pricing Nth-to-Default CDS

Pricing a Nth-to-Default (NTD) CDS requires combining the marginal default probabilities of the underlying bonds with their joint dependence structure to determine the probability that the n-th default occurs by a given time.

Let τ_n denote the time of the n-th default in a basket of N bonds. The expected protection leg cash flows are computed by weighting the loss given default by the probability that the n-th default occurs in each time interval. Mathematically, this involves summing over all combinations of n bonds that could default first and computing the joint probability that those n bonds default by a given time while the remaining $N - n$ bonds survive. These probabilities are obtained from the joint survival distribution, either via a copula model or a correlated Monte Carlo simulation.

The premium leg is similarly computed by discounting periodic payments weighted by the probability that fewer than n defaults have occurred. Integrating these probabilities over time yields the present values of the protection and premium legs, from which the par spread for the NTD CDS can be derived.

For the n-th to default:

$$\tau_n = \inf\{t : \text{at least } n \text{ names defaulted}\}.$$

Conditional survival probabilities are combined combinatorially:

$$\mathbb{P}(\tau_{(2)} > t \mid m) = \sum_{i<j} \bar{Q}_i(t \mid m).\bar{Q}_j(t \mid m) \prod_{k \neq i,j} Q_k(t \mid m),$$

and similarly for general n. Integration over M yields unconditional probabilities.

7 Correlation

Accurate pricing of Nth-to-Default (NTD) CDS relies critically on the correlation parameter linking each bond to the latent systemic credit factor. Since direct pairwise correlations between individual bonds are rarely observable, market participants typically calibrate the correlation from **liquid credit index tranches**, which provide transparent pricing and implied correlation data. These tranches serve as a practical proxy for the joint default behavior of the underlying basket and allow the one-factor Gaussian copula model to remain consistent with market quotes.

- For North American credit-referenced bonds, the **CDX equity tranche 0–3%** is often used as the primary reference for calibration.
- The one-factor Gaussian copula framework implies an **equi-correlation matrix**, where the off-diagonal elements are set to the implied correlation ρ and the diagonal elements are 1.
- Market prices and the corresponding implied correlations are typically obtained from Bloomberg, using the **CDOT** (Credit Default Option Tranche) and **CTRU** (Credit Tranche) pages.

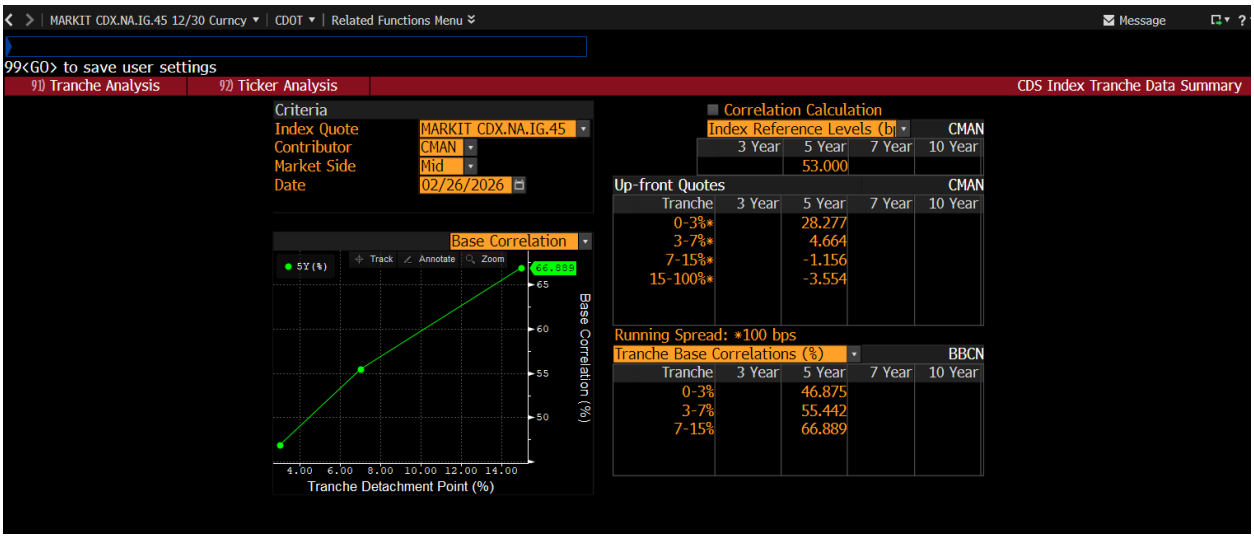

Figure 1: Bloomberg CDOT: Implied Correlation Quotes

7.1 Correlation Profiles

7.1.1 First-to-Default CDS

Buying protection on a First-to-Default CDS exposes the investor to the risk of early default in the basket, which is highly sensitive to default correlation. In this context, the position is effectively **short correlation**, because higher correlation reduces the probability of a first default occurring early, while lower correlation increases it.

Highest Par Spread: When the default correlation between the bonds is zero, defaults are independent. The probability of at least one bond defaulting early is maximized, so the par spread of the FTD CDS approaches the **sum of the individual single-name credit spreads**.

Lowest Par Spread: When the correlation is one, all bonds move perfectly together. Defaults tend to cluster, so the first default occurs no earlier than the weakest (highest risk) bond would default on its own. In this case, the par spread of the FTD CDS equals the **highest single-name credit spread in the basket**.

7.1.2 Nth-to-Default CDS (n¿¿1)

Buying protection on a high Nth-to-Default CDS exposes the investor to the risk that multiple defaults occur in the basket. In this context, the position is effectively long correlation, because higher correlation increases the likelihood that enough defaults occur simultaneously to trigger the Nth-to-Default payoff, while lower correlation decreases it.

Highest Par Spread: When the correlation between the underlying bonds is one, defaults move together perfectly. The probability that the Nth default occurs is maximized, so the par spread approaches the highest single-name credit spread in the basket.

Lowest Par Spread: When the correlation is zero, defaults are independent and clustering is unlikely. The probability of N or more defaults occurring early is very low, so the par spread is

Figure 2: Bloomberg CTRU: Tranche Pricing

close to zero.

8 Risk

Key Risk Exposures for Nth-to-Default CDS

- **Correlation Delta**: measures the sensitivity of the NTD CDS present value to changes in the implied correlation between the underlying bonds. For FTD CDS, this is typically negative (short correlation), whereas for high Nth-to-Default, it is positive (long correlation). Monitoring correlation delta is essential for managing the nonlinear dependence of the position on default clustering.

$$Delta_\rho = \frac{\partial \mathrm{PV}}{\partial \rho}$$

- **CS01**: the sensitivity of the NTD CDS to a 1 basis point (0.01%) change in the credit spread of a single-name CDS within the basket. CS01 hedging is the primary method for offsetting single-name spread risk and isolating the correlation-driven component of the position.

$$\mathrm{CS01}_i = \frac{\partial \mathrm{PV}}{\partial s_i}$$

- **Cross Gamma**: measures the interaction between changes in single-name spreads and correlation. It captures the nonlinear effect where shifts in spreads can amplify or dampen correlation exposure, and vice versa. Cross gamma is particularly important in stressed market conditions where both spreads and correlations can move significantly, creating non-intuitive P&L impacts.

$$\Gamma_{i,\rho} = \frac{\partial^2 \mathrm{PV}}{\partial s_i \, \partial \rho},$$

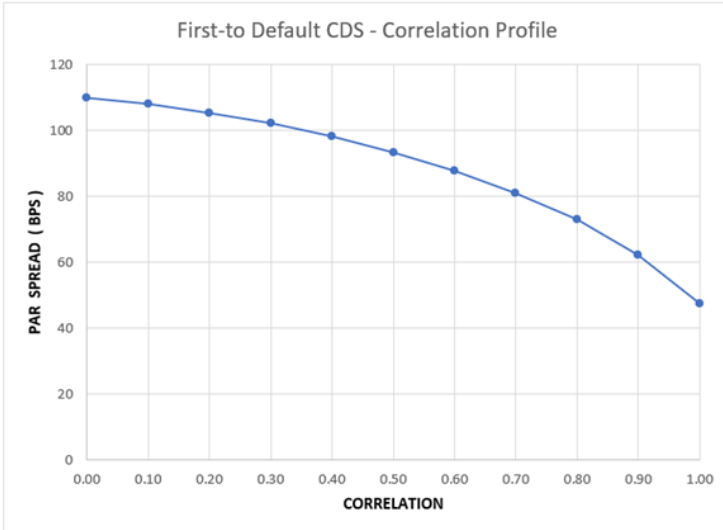

Figure 3: First-to-Default CDS: Par Spread vs Implied Correlation

9 Hedging

By using **single-name CDS to hedge the CS01 exposure** of an FTD or NTD position, a trader can effectively neutralize the sensitivity of the basket to individual credit spreads. Each hedge is sized according to the ratio of the single-name CS01 to the total basket CS01, as indicated in the formula above. Once the CS01 hedges are in place, the residual risk of the NTD or FTD CDS is dominated by **correlation exposure**. This isolates the position's dependence on the joint default behavior of the basket, allowing the trader to express a view purely on default clustering and systemic credit risk. The remaining correlation risk is then monitored and managed through **correlation delta** (Δ_ρ) and **cross gamma** ($\Gamma_{i,\rho}$) metrics, providing a clean separation between spread-driven and correlation-driven P&L.

$$N_{\text{Hedge}_\text{i}} = -N_{\text{Basket}} \cdot \frac{\text{CS01}_i}{\text{CS01}_{\text{Basket}}},$$

10 Gap Risk

Gap risk arises when multiple risk factors move sharply and simultaneously, creating sudden and large P&L changes that are difficult to manage. In the context of Nth-to-Default CDS, this typically occurs when **credit spreads widen rapidly** while **default correlations spike**. Such simultaneous moves increase the likelihood that multiple defaults occur in a short time frame, amplifying losses beyond what would be expected from single-name spread moves alone.

Distressed market conditions often exacerbate this risk because **liquidity deteriorates**, making it harder to adjust hedge positions or exit trades quickly. Even well-hedged CS01 exposures can leave a portfolio vulnerable, as correlation-driven losses may materialize faster than positions can be rebalanced.

To mitigate gap risk, it is essential to **monitor position sizes**, use stress-testing techniques, and track metrics such as **Value at Risk (VaR)** and **Expected Shortfall**, ensuring that sudden market dislocations do not generate outsized, unmanaged exposures.

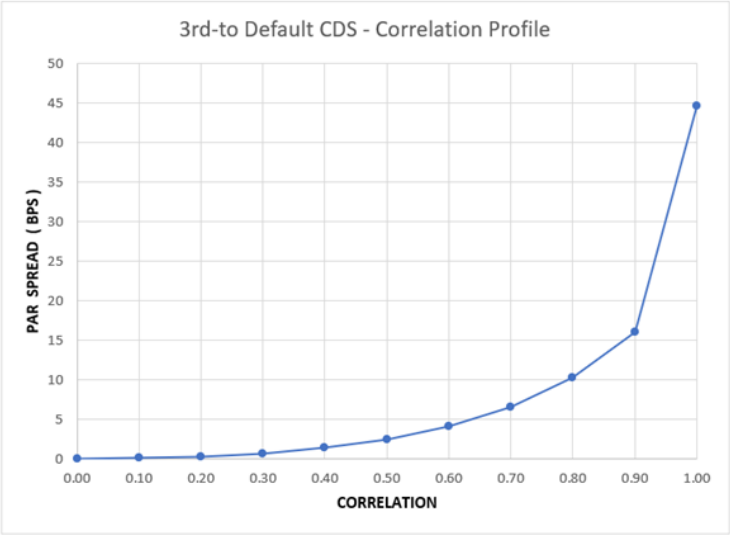

Figure 4: Third-to-Default CDS: Par Spread vs Implied Correlation

11 Conclusion

Nth-to-Default (NTD) CDS provide **targeted exposure to the clustering of credit events** within a bond portfolio, making them uniquely sensitive to both single-name credit spreads and default correlation. These instruments are actively traded for **yield enhancement**, as the par spreads of baskets can exceed individual credit spreads, and for **correlation trading**, allowing investors to express a view on the joint default behavior of multiple issuers.

Market-standard valuation is typically performed using the **one-factor Gaussian copula framework**, which links each bond's default to a latent systemic credit factor while remaining consistent with observed single-name CDS spreads. Analytical techniques, such as numerical integration over the systemic factor, are commonly complemented by **Monte Carlo simulation** to validate results or handle more complex basket structures. The dependence parameter, interpreted as an implied correlation, is usually calibrated from **liquid credit index tranches** such as CDX in North America or iTraxx in Europe, ensuring consistency with observable market data.

Because NTD CDS exhibit **nonlinear sensitivities** to both spreads and correlation, careful risk management is essential. CS01 hedging using single-name CDS can isolate correlation exposure, which is then monitored via **correlation delta** and **cross gamma**. Gap risk, which arises when widening spreads, spiking correlations, and declining liquidity coincide, further underscores the importance of stress testing, position sizing, and continuous monitoring.

In summary, NTD CDS offer traders and portfolio managers a powerful tool to extract yield, hedge portfolio exposures, and take deliberate positions on credit clustering and correlation. Mastery of both pricing models and risk techniques is necessary to trade these products effectively and manage the complex interplay of spreads, correlation, and liquidity risk.

12 Recommended Reading

For further study on the quantitative foundations and practical implementation of credit derivatives and the Gaussian copula model, the following references are highly recommended:

- Li, David X., *"On Default Correlation: A Copula Function Approach"*, **Journal of Fixed Income**, 2000.
- Hull, John C., *"Options, Futures, and Other Derivatives"*, 11th Edition, Pearson, 2020.
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- O'Kane, David, *"Modelling Single-name and Multi-name Credit Derivatives"*, Wiley, 2008.
- S&P Dow Jones Indices, *CDS Indices Primer*, a foundational introduction to credit default swap indices. Available at: https://www.spglobal.com/spdji/en/documents/campaigns/cds-indices-primer.pdf